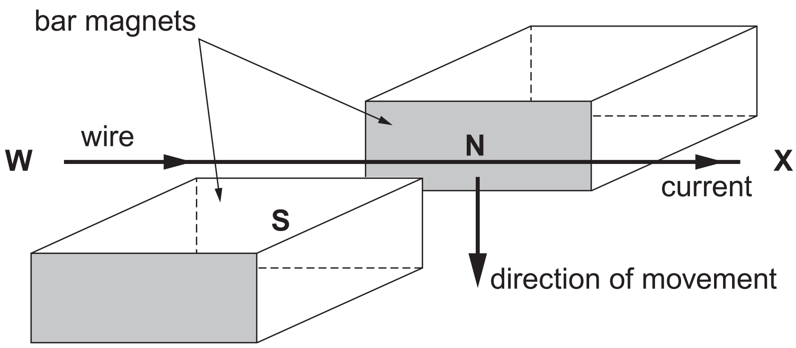
Examiner only

2. A teacher places a wire in a magnetic field. A current is passed through the wire and a force makes it move downwards. The following diagram shows how the apparatus was set up to show this effect.



A current is passed through the wire from **W** to **X**. A force makes the wire move **downwards**.

(a) Tick (✓) the boxes alongside the **three** correct statements below. [3]

The direction of the magnetic field is from N to S in the diagram.	☐
The direction of movement is given by Fleming's right hand rule.	☐
A current from **X** to **W** makes the wire move upwards.	☐
When two South poles face each other the wire moves upwards.	☐
A stronger magnetic field makes the wire move upwards.	☐
This effect is used in electric motors.	☐

(b) A student suggests to the teacher that the effect could be investigated further. Explain how the teacher could make changes that would affect the **size** of the force on the wire. [3]

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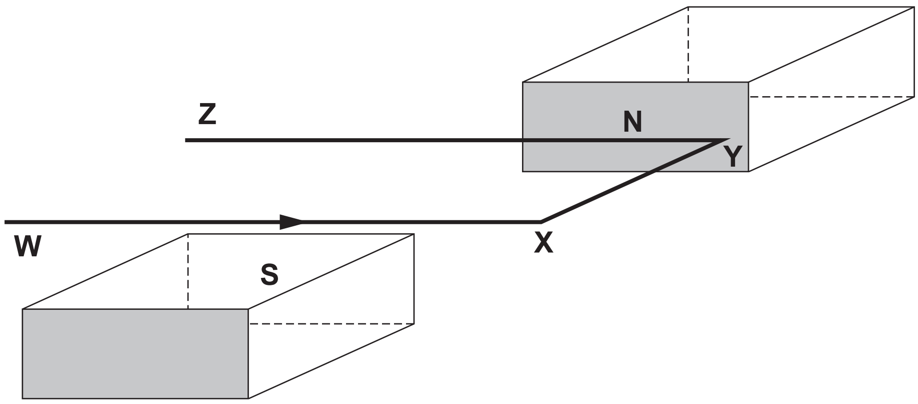
Examiner only

(c) In another experiment using the magnets, a single coil of wire **WXYZ** is placed between the magnets. A current is passed through the coil in the direction shown by the arrow.



Draw an arrow on the diagram to show the direction of the current in **YZ**.
Label the arrow C. [1]

Draw a second arrow on the diagram to show the direction of the force on **YZ**.
Label the arrow F. [1]

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